SOHAN GADEWAR

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Education

Vishwakarma Institute of Information Technology **2023 – 2027**
B.Tech in Information Technology | ***CGPA: 8.97/10.00*** *Pune, India*
Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, System Design, Object Oriented Programming, Machine Learning.

Experience & Leadership

India Innovates Hackathon **Jan 2026 – Feb 2026**
National Finalist & Lead Builder — Ministry of Education & AICTE *Pune, India*
• Ranked among **top 20% finalist teams** nationwide out of **6,000+ participating teams** across India; spearheaded product discovery, scoping, and rapid cross-functional sprints.

GHCI 25 National GenAI Hackathon **2025**
Round 2 Qualifier — AnitaB.org India & Backbase ("Unbound with GenAI") | *Credential Certificate* *National Hackathon*
• Selected for Round 2 in national GenAI challenge; engineered agent workflows and automated evaluation harnesses with structured JSON guardrails, validating prototype reliability under strict prompt constraints.

South African Patent Office (Filed Patent) **2025**
Patent Applicant & Inventor — AI-Powered Lost & Found Matching System *Status: Patent Filed*
• Conceptualized and filed patent for an AI matching architecture utilizing a **5-factor blind multimodal scoring engine** (image embeddings, NLP text semantics, spatio-temporal telemetry) with automated urgency detection.

Projects

1. Project Sentinel: AI Decision Memory & State-Drift Agent **Aug 2026 – Sept 2026**
Python, PostgreSQL, FastAPI, Lovable | Live Prototype | Notion PRD *Enterprise AI System*
• Spearheaded product discovery across engineering standups, Slack, and Jira, identifying critical context loss and formulating the **"Silent-by-Default"** thesis to eliminate unprompted bot summaries and protect developer attention.
• Architected deterministic audit engine powered by **append-only PostgreSQL schemas**, surfacing discrepancies between spoken commitments and recorded codebase state strictly upon explicit user queries.
• Shipped clickable Lovable MVP with live drift queries; authored comprehensive PRD detailing state transition matrices, query grammar, and phased rollout while scoping out employee surveillance features.

2. CausalRetention: Prescriptive Uplift & Customer Retention Engine **Oct 2026**
Python, XGBoost, Streamlit, Docker | Live Decision Suite | CATE Spec *Causal ML Engine*
• Uncovered key economic friction in retention marketing where conventional churn models incur a **-13.1% net ROI** due to deadweight spend on non-responsive "Lost Causes" and churn acceleration from nudging dormant "Sleeping Dogs".
• Architected prescriptive AI decision engine using a **Two-Model (T-Learner) XGBoost pipeline** to calculate Conditional Average Treatment Effect (CATE), segmenting customer cohorts into 4 distinct quadrants.
• Engineered a constrained **Knapsack financial budget optimizer** under a fixed $15K constraint, capturing **+$112.7K+ in net preserved revenue (+752% campaign ROI)**; validated incrementality via Qini curves in containerized Streamlit suite.

3. FlashDrop: High-Concurrency Flash Release & ACID Inventory Engine **Oct 2026**
Java, Spring Boot, MySQL, HikariCP | GitHub Repository & Benchmarks *High-Concurrency Backend*
• Architected a high-concurrency flash release inventory engine in Spring Boot backed by MySQL 8.0, eliminating race conditions and guaranteeing **zero overselling** via InnoDB pessimistic row-level locking (**SELECT ... FOR UPDATE**).
• Designed a **3NF relational schema with ACID stored procedures** (sp_reserve_product, sp_checkout_order), ensuring atomic multi-table state transitions across limited inventory, buyer holds, and finalized orders.
• Configured optimized **HikariCP connection pooling** and automated native **MySQL EVENT SCHEDULER** routines to reclaim expired 5-minute temporary holds; verified 0% data inconsistencies via a multi-threaded Python stress suite.

Technical Skills

Languages: Python, Java, SQL, Golang, C++, C, JavaScript
Frameworks: Spring Boot, FastAPI, Streamlit, XGBoost, Scikit-learn, React
Databases & Cloud: PostgreSQL, MySQL, Supabase, Docker, Git, Linux
Product & Tools: PRD Writing, Jobs-to-be-Done (JTBD), RICE Prioritization, North Star & Counter-Metrics, A/B Testing, Figma, Postman, Notion

Certifications

• **Product Management:** Product Management Masterclass (GeeksforGeeks) — Product lifecycle, JTBD discovery frameworks, and roadmap metrics.
• **AI & Workflows:** Prompt Engineering | Advanced RAG (CampusX) | Docker for ML (CampusX) | n8n Automation (CampusX)